PURCHASE DESCRIPTION FOR THE ACQUISITION OF ONE (1) VERTICAL TURNING CENTER

1.0 SCOPE.

1.1 Scope. The content of this specification describes the minimum requirements of the government for one (1) CNC Vertical Turning Center (VTC) onboard MCAS Cherry Point. It is the government's intent that a single (primary) contractor be awarded this contract and be responsible for the accomplishment of all work detailed by this specification including but not limited to the provisioning of equipment, removal of existing equipment, demolition, new foundation, and installation.

2.0 APPLICABLE DOCUMENTS.

2.1 General. The documents listed in this section are cited in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed in section 2. Unless otherwise specified, the documents listed in the solicitation or contract shall be the current edition at time of solicitation.

2.2 Government Documents.

2.2.1 Specifications, Standards, and Handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the solicitation or contract.

STANDARDS – FEDERAL

FED-STD-H28B: Screw Thread Standards for Federal Service

U.S Department of Labor, Occupational Safety and Health Administration (OSHA)

29 CFR 1910 Vol. 5: General Industry, OSHA Safety and Health Standards

(Application for copies should be addressed to the OSHA Publications Superintendent of Documents P.O. Box 37535 Washington, D.C. 20013-7535)

Environmental Protection Agency (EPA)

40 CFR 761: Polychlorinated Biphenyls (PCBs) Manufacturing, Processing, Distribution in Commerce, and Use Prohibitions

(Application for copies should be addressed to the EPA Ariel Rios Building 1200 Pennsylvania Avenue, N.W. Washington, DC 20460)

MILITARY STANDARDS

MIL-STD-130N w/Change 1: Identification Marking of U.S. Military Property Item Unique Identification Marking (IUID)

(Application for copies should be addressed to Air Force Product Data Systems Modernization Standards (TMSS) Office, 554 ELSG/SBP, 4375 Chidlaw Road, Bldg. 262, Room S008, Wright-Patterson AFB, OH 45433-5006, Phone (937) 257-0870 or 0853 or https://quicksearch.dla.mil/qsSearch.aspx)

2.2.2 Other Government Documents, Drawings, and Publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the solicitation or contract.

LOCAL FACILITY DOCUMENTS

RESERVED

(Applications for these documents should be addressed to the contracting officer)

2.3 Non-Government Publications, Specifications, Standards, and Handbooks. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the solicitation or contract.

American Gear Manufacturers Association (AGMA)

AGMA ISO 1328-1-B14: Cylindrical Gears - ISO System of Flank Tolerance Classification - Part 1: Definitions and Allowable Values of Deviations Relevant to Flanks of Gear Teeth

(Address application for copies to: 500 Montgomery Street, Suite 350, Alexandria, VA 22314-1581 USA)

American National Standards Institute, Inc. (ANSI)

ANSI B5.1M: T-Slots, their Bolts, Nuts, and Tongues, Revision of ANSI B5.1-1975

B11/STD B11: General Safety Requirements Common to ANSI B11 Machines

B11/STD B11.22: Safety Requirements for Turning Centers and Automatic, Numerically Controlled Turning Machines

(Address application for copies to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036)

American Society of Mechanical Engineers (ASME)

ASME B5.48: Ball Screws

ASME Y14.5 Geometric Dimensioning and Tolerancing

ASME Y14.100: Engineering Drawing Practices

(Applications for copies should be addressed to ASME International Three Park Avenue New York, NY 10016-5990 or www.asme.org)

International Organization for Standardization (ISO)

ISO 13041-4: Test Conditions for Numerically Controlled Turning Machines and Turning Centers - Part 4: Accuracy and Repeatability of Positioning of Linear and Rotary Axes.

ISO 13041-6: Test Conditions for Numerically Controlled Turning Machines and Turning Centers - Part 6: Accuracy of a Finished Test Piece

ISO 230-2: Test Code for Machine Tools – Part 2: Determination of Accuracy and Repeatability of Positioning of Numerically Controlled Axes

ISO 230-7: Test Code for Machine Tools – Part 7: Geometric Accuracy of Axes of Rotation

ISO 4413: Hydraulic fluid power-General rules and safety requirements for systems and their components

ISO 4414: Pneumatic power-General rules and safety requirements for systems and their components

(Application for copies should be addressed to the International Organization for Standardization (ISO) 1 rue de Varembé, Casepostale56, CH211Geneva 20, Switzerland)

National Electrical Manufacturers Association (NEMA)

NEMA 250: Enclosures for Electrical Equipment (1000 V Maximum)

NEMA ICS 16: Industrial Control and Systems: Motion/Position Control Motors, Controls, and Feedback Devices

NEMA MG 1: Motors and Generators

NEMA Z535.4: Product Safety Signs and Labels

(Application for copies should be addressed to the National Electrical Manufacturers Association, 1300 N 17th Street, Rosslyn, VA 22209)

National Fire Protection Association (NFPA)

NFPA 70: National Electrical Code

NFPA 79: Electrical Standard for Industrial Machinery

NFPA 484 Standard Development

NFPA 652 Dust Hazard Analysis

(Applications for copies should be addressed to the National Fire Protection Association, ATTN: Support Services, 11 Tracy Drive, Avon, MA 02322)

Society of Automotive Engineers (SAE)

SAE EIA-274-D: Interchangeable variable block data contouring, format for positioning and contouring/positioning numerically controlled machines

(Application for copies should be addressed to: Society of Automotive Engineers 400 Commonwealth Drive, Warrendale PA 15096)

2.4 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulation unless a specific exemption has been obtained.

2.5 Acronym Definitions Not Explicitly Listed in Text. The following acronyms not explicitly listed in each section text are defined as follows:

a. JIS- Japanese Industrial Standards
b. DIN- German Institute for Standardization
c. EN- European Standard
d. IEC – International Electrotechnical Commission

2.6 Proposal Format. Proposal shall be submitted using the following mathematical conventions:

a. Whole numbers will be expressed where one thousand is written as 1,000.
b. Decimals will be expressed where one-tenth is written as 0.1.

3.0 REQUIREMENTS.

3.1 Design. The item offered shall be new and a current commercially available model. A current model is defined as the manufacturer's currently produced model which, on the date this solicitation is issued, has been designed, engineered, sold or is being offered for sale through advertisements or manufacturer's published catalogue or brochures. Products such as a prototype unit, pre-production model, or experimental unit do not qualify as meeting the requirements specified herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage or distortion shall be accessible for adjustment, replacement, and repair.

3.1.1 Measuring and Indicating Device Graduations. All measuring and indicating devices on the machine shall be graduated in the United States customary units. An electronic control system that demonstrates switching between manufacturer standard units to United States customary units will be found acceptable.

3.1.2 Controls. All electrical, mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's workstation(s).

3.1.3 Safety and Health Requirements. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 29 CFR 1910. Safety signs and labels on all equipment shall comply with NEMA Z535.4.

3.1.3.1 Safety Guarding. Covers, guards, or other safety devices shall be provided per ANSI B11 and B11.22 or equivalent ISO/EN/DIN/JIS standards. At the time of proposal, the vendor shall indicate which document they will utilize for compliance to the requirements. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be easily removed to facilitate inspection, maintenance, and repair of the parts.

3.1.4 Audible Noise Level. Audible noise emitted by the equipment shall not exceed 84 decibels (dB), measured on the "A" weighted scale of a standard Type II sound level meter, at the operator's work position or any point at a distance of three (3) feet from the equipment. Noise generated by the work piece shall be excluded in determining compliance of the equipment with the 84 dB requirements.

3.1.5 Hazardous Material Exclusions. Supplies or materials being provided as part of the equipment shall be free of known hazardous materials. These materials shall not be brought on site without prior approval of the Government. Radioactive materials or instruments capable of producing ionizing radiation as well as materials which contain asbestos, mercury, lithium, methylene chloride, lead (= or >0.06%), or polychlorinated biphenyls (PCB's) are prohibited. Definitions of hazardous materials are specified in the latest version, including revisions adopted during the term of the contract, of FED-STD-313F. Nickel Metal Hydride and lithium batteries are permissible for memory backup. Class I Ozone Depleting Substances as defined in 40 CFR 82 shall not be used in the performance of this contract or be provided as part of the equipment. Exceptions to the prohibition of these materials must be referred to the Contracting Officer in writing, for consideration, not later than 60 calendar days after effective date of contract and prior to any work being performed.

3.1.6 **Environmental Protection.** Under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.1.7 **Lubrication.** Means shall be provided to ensure proper automatic lubrication for all moving parts in accordance with manufacturer requirements. All oil holes, grease fittings, and filler caps shall be easily accessible.

3.1.7.1 **Lubrication Data.** The Contractor shall provide the following information on all liquid or grease lubricants contained within equipment/system or in bulk (5 gallons or more) supplied under this procurement.

a. Liquid Lubricants
   i. Viscosity grade in ISO units
   ii. Viscosity Index
   iii. AGMA and/or SAE classification as applicable
   iv. Mil Spec and/or NATO Code as applicable
   v. Viscosity in Centistokes (cst) @ 40° C and 100° C
   vi. Type Additive package. Example: ZDDP, Rust inhibitors, Foam inhibitors.
   vii. Neutralization, the TBN (alkalinity) and/or TAN (acidity).
   viii. Flash point
   ix. Pour point
b. Grease Lubricants
   i. National Lubrication and Grease Institute (NLGI) Number
   ii. Type and percent of thickener
   iii. Dropping point
   iv. Maximum Service Temperature
   v. Base oil viscosity range in SUS or centipoise

3.1.7.2 **Recirculating Lubrication.** When the machine is equipped with a recirculating lubrication system, it shall include a filter that is cleanable or replaceable. Each lubricant reservoir shall have at least a 24-hour capacity. Means shall be provided to indicate a low lubrication condition. When a low lubrication condition occurs, the machine shall complete the current operation and come to a complete stop. The machine shall indicate that lubrication is required.

3.1.8 **Replacement Parts.** All parts subject to replacement shall be manufactured to definite dimensions and tolerances. Replacement parts shall be interchangeable without requiring modification. Replacement parts are those that are subject to wear or failure such as gears, bearings, shafts, feed screws, drive belts, pumps, and seals. Electrical replacement parts shall include but not be limited to, motors, relays, fuses, switches, magnetics, pushbuttons, light bulbs, braking mechanisms, and solid-state devices. Tooling and accessories normally supplied as standard items shall also be considered replacement parts. Foreign-sourced replacement parts shall either be interchangeable with domestically supplied components or be readily available from a domestic source.

3.2 **<u>Construction.</u>** The machine shall be constructed of parts that are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without distortion.

3.2.1 **<u>Castings and Forgings.</u>** All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, filling with solder or paste shall be used for reclaiming any defective part.

3.2.2 **<u>Fastening Devices.</u>** All screws, pins, bolts, and other fasteners shall be installed to prevent unintentional loosening. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.2.3 **<u>Threads.</u>** All threaded parts used on the machine and related attachments and accessories shall conform to FED-STD-H28B and the applicable "Detailed Standard" section referenced therein or equivalent ISO/EN/DIN/JIS standards. At the time of proposal, the vendor shall indicate which document they will utilize for compliance to the requirements.

3.2.4 **<u>Hydraulic System.</u>** Under the condition that a machine has a hydraulic system, the system shall conform to ISO 4413. The system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. A machine that does not require a hydraulic system to operate properly shall be considered acceptable.

3.2.5 **<u>Pneumatic System.</u>** Under the condition that a machine has a pneumatic system, the system shall conform to ISO 4414. The system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. A machine that does not require a pneumatic system to operate properly shall be considered acceptable.

3.2.6 **<u>Gears.</u>** All gears of axis drive trains shall conform to or exceed all provisions of ANSI/AGMA ISO 1328-1-B14. All gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in drive trains shall be hardened and ground steel.

3.2.7 **<u>Surfaces.</u>** All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. The condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.2.8 **<u>Welding, Brazing, or Soldering.</u>** Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.2.9 **<u>Painting.</u>** The machine shall be painted in accordance with the manufacturer's commercial practice. All paint shall be of the lead free type.

3.2.10 **Workmanship.** Workmanship of the machine shall meet all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this purchase description.

3.3 **Components.** The machine shall consist of not less than the following components:

3.3.1 **Base.** The base shall be an iron or iron alloy casting or weld element which is fully stress relieved, and webbed and braced to minimize distortion and deflection and shall possess the mass, strength, rigidity and other load carrying characteristics necessary for supporting the machine column, cross rail, saddle, ram, turning spindle, and their associated components. It shall be constructed to maintain mutual component alignment for assuring the accuracy required herein. The base surfaces that create the machining enclosure floor shall be sloped towards the chip conveyors to direct chips and coolant to the chip conveyors. Means shall be provided for securing and leveling the machine to its foundation. The base shall have access or removable plates as necessary for servicing components housed within.

3.3.2 **Column.** The column shall be an iron or iron alloy casting or weld element which is fully stress relieved, and webbed and braced to minimize distortion and deflection under normal tool loads. The column shall be securely fastened and aligned to the machine base, forming the foundation for an elevating cross rail and its associated drive components. Scraping the base to column mounting surface shall be used to achieve the geometrical tolerances requirements IAW Table II. Jacking screws are not an acceptable means of achieving geometry. Vertical box type guide ways shall be provided to support and guide an elevating cross rail assembly within its full rated travel and be IAW Table I.

3.3.3 **Table.** The worktable shall be stationary in the X and Y-axes with rotation about the C-axis. The table shall meet the size and performance characteristics noted in 3.4 Table I. The table shall have bi-directional control to provide for virtual Y axis capabilities. The worktable shall be a casting of iron, or iron alloy, internally ribbed and proportioned to support and withstand the maximum capacity work-piece weights and tool thrust loads. The worktable shall be precision fit to the bearing system. The bearing system shall be able to accept both radial and axial loads. The top of the worktable shall have a machined or ground finished for a smooth work-mounting surface. The table shall support a track system of four (4) tracks at a width of 3.25 inches separated at 90 degrees that accept manual chuck jaws. This chuck jaw system shall be robust and shall not require any calibration or automated systems to operate. A set of jaws shall be provided with the machine. The table shall also include T slots in accordance with ANSI B5.1M or current equivalent ISO/EN/DIN/JIS/IEC standards. At the time of proposal, the vendor shall indicate which document they will utilize for compliance to the requirements. There shall be 4 slots at a radial length 27.5 inches centered forty-five (45) degrees from the manual jaw tracks. There should be eight (8) T-slots at a radial length 20.5 inches at 15 degrees on either side of the four (4) longer T-slots. There shall be a blind hole located at the table center at a minimum diameter of 2.5 inches.

3.3.4 **Axes.** All drive and feed motors shall be of the Fanuc brand to match the controller specified in section 3.3.15. All linear axes of the machine shall be actuated through a closed loop, high-gain electrical servo mechanism driving re-circulating ball screws and nuts. Infinitely variable feed drives shall be provided for all axes. Axes shall be machined to accept way bearings of a precise fit to their respective ways. The way bearings shall be an antifriction type and shall be provided with adjustable gibs, guides, or bearings to maintain and adjust alignment of the table to the ways and prevent causing excessive loading of the feed drive motor. Antifriction, re-circulating ball type linear ways are acceptable. All motion/position control motors, controls, and feedback devices shall conform to NEMA ICS 16 or current equivalent ISO/EN/DIN/JIS/IEC standards. At the time of proposal, the vendor shall indicate which document they will utilize for compliance to the requirements. Axis description shall be as follows:

CNC Controlled:
X-axis – Horizontal ram travel on the crossrail (hydrostatic guideways)
Z-axis – Vertical ram travel (hydrostatic guideways)
C-axis – Bidirectional table rotation
W-axis – Crossrail vertical movement, indexed incrementally

3.3.5 Machine Ways. Machine ways shall be hardened. The ways shall be of a type suitable for high-grade precision numerically controlled machines covered by this purchase description. Movement of the slides on the way surfaces shall be through adjustable preloaded, re-circulating roller bearing cartridges or nonmetallic way liners having a low coefficient of friction, with excellent slip-stick and damping properties. Protection of the machine ways shall be either telescoping or curtain type way covers equipped with way wipers.

3.3.6 Cross-Rail. The cross rail shall be a continuous one-piece iron, or iron alloy, casting or weld element which is fully stress relieved, webbed and braced and proportioned to support and withstand the ram assembly and tool thrust loads. Cables, hoses and conductors attached to the elevating cross rail shall be mounted in a location that provides an interference free environment for the elevating cross rail and the normal work envelope. The elevating cross rail shall utilize column mounted boxed guide ways for linear travel accuracy and rigidity in the W-axis. The cross rail movement shall utilize a positioning stop of four (4) equally spaced locations along the W-axis travel. The elevating cross rail assembly shall consist of motors and accessories necessary to hydrostatically support and provide the ram head assembly with programmable horizontal travel about the X-axis. Cross-rail controls shall provide a fully programable positioning axis with dual scale feedback system. The steel cross rails shall be hardened and precision ground. The cross rails shall be enclosed with way covers for protection against chips, dirt, coolant and other abrasive materials. Cross-rail shall be provided with an automatic leveling and clamping system that maximizes rigidity. If required, cross-rail controls/travel shall prevent any crashes with guarding.

3.3.7 Ram. The machine shall consist of a single column forged ram. The ram guideways shall be hardened, precision ground and be of the hydrostatic type to minimize load transfer and vibration excitation. The Ram shall be supported by continuous solid one-piece iron or iron alloy cast housing. The one-piece casting shall be affixed to the cross rail. Ram shall be supported and held in axial and radial alignment with fully enclosed ram with a minimum cross-sectional area of 90 square inches. The ram shall be capable of full travel through the X-axis. Drilling and milling operations shall be achieved in the virtual Y-axis via a live spindle attachment. Ram shall be configured to accept both standard, square shank turning tools as well as a vertical milling head including all the tooling defined in Section 3.6.1 utilizing an automatic clamping system.

3.3.8 Live Milling Spindle Head. A Capto C8 tool-holding interface spindle, or equivalent high-performance, modular, quick-change tooling system spindle designed for heavy-duty turning, milling, and drilling shall be provided. The spindle shall be supported and held in axial and radial alignment. Integrated spindle monitoring shall monitor vibration, unbalance, and bearings. Spindle bearings shall be lubricated by the air oil-mist lubrication method or with permanently greased bearings.

3.3.9 Machine Enclosure. Full enclosure shall incorporate two front viewing windows and interior camera for monitoring the workspace. Camera system shall be capable of recording and providing live feed to the controller. Full enclosure shall be flush floor mount style allowing access to the work area. Full enclosure doors shall allow for crane loading of parts.

3.3.10 Coolant System. A flood type coolant system shall be provided. The coolant system shall include a sump or reservoir, a power-driven pump, and all necessary piping. The system shall have means for draining and cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant over the entire work area. The coolant system shall be compatible with HOCUT 795B type coolant. Coolant shall be delivered through the ram to the tool during cutting operations. Through coolant shall be turned on and off at the controller. The coolant system shall be supplied with a filtration system to keep the coolant system free of particulates and unclogged to operate at maximum flow. A fully functioning chip wash gun shall be provided near the operator's work's station with reach to the extents of the work zone. The wash gun shall utilize the coolant from supply tank of machine. An oil skimmer option shall be included to remove any oil accumulations of oil in the top of the coolant tank.

3.3.10.1 Oil Chiller. This machine shall contain a hydrostatic oil chiller that supports the machine's working envelope.

3.3.11 Chip, Dust, and Coolant collection. This machine shall be able to collect mist and chips in both dry and wetting cutting operations. It is acceptable to support this requirement with two different collection units, but all functions must be accessible at the controller. Coolant system shall include coolant pan with a conveyor system including a hopper for chip collection. Chips collected from the conveyor shall be deposited in a sealed 55-gallon drum. This coolant pan is to be furnished with a cover to form a barrier when machining Magnesium. Dry metal dust collection unit shall be in accordance with NFPA 484 and NFPA 652.

3.3.12 Automatic Tool Changer. A carousel type Automatic Tool Changer (ATC) shall be provided to hold heads/accessories and modular tools. This Tool changer shall support fifteen (15) tool positions. This tool changer shall have its own mounting/support system and not be mounted to the vertical turning center base itself.

3.3.13 Part Probe. The machine shall be provided with a part probe. The probe shall measure inside and outside diameters of 1 inch (minimum) and the full length of the Z-axis travel. The probe's accuracy and repeatability shall be equal to the machine's capabilities stated in section 3.5 Table II. The probe data shall be capable of being copied to the hard drive and a printer. The probe shall be capable of being inserted directly into the ram set up.

3.3.13.1 Tool Probe. A tool probe shall be provided and have initial settings for all cutting tools and referencing back to the datum surface. The tool probe shall also have field calibration at regular intervals. The tool probe shall be contact type. The machine shall be provided with an inductive pick-up style part probe. The probe shall measure inside and outside diameters of 1 inch (minimum) and the full length of the Z-axis travel. The probe's accuracy and repeatability shall be equal to the machine's capabilities stated in section 3.5 Table II. The probe data shall be capable of being copied to the hard drive and a printer. The probe shall be capable of being inserted directly into the Ram set up.

3.3.14 Electrical System. The machine shall conform to NFPA 79 or current equivalent ISO/EN/DIN/JIS/IEC standards. At the time of proposal, the vendor shall indicate which document they will utilize for compliance to the requirements. The existing source power available at the facility is 480 volt, 3-phase, 60 Hertz (Hz). The machine's electrical system shall be tolerant of fluctuation of ± 10 percent. The electrical system shall be complete, including any electrical transformer(s) that may be required to modify the existing source voltage to the proper operating voltage of the equipment. A properly rated, and fused, single disconnect device shall be utilized on the machine with means of lockout in the off position only.

3.3.14.1 Electrical Enclosures. Construction of the electrical enclosures shall be in accordance with NEMA 250 type 12 or current equivalent ISO/EN/DIN/JIS/IEC standards and shall provide for ventilation of components and shall be designed for an indoor non-hazardous location. All electrical components shall conform to NFPA 79 or equivalent ISO/EN/DIN/JIS/ UL508 standards. The machine shall be provided with interlocks on its electronic cabinet that shall instantly remove power from the equipment when the door is opened in accordance with NFPA 79 or current equivalent ISO/EN/DIN/JIS/IEC/UL508A standards. At the time of proposal, the vendor shall indicate which document they will utilize for compliance to the requirements.

3.3.14.2 Arc Flash Protection. The machine enclosures shall be marked for arc flash in accordance with NFPA 79 Paragraph 16.2.3 "Safety Signs for Electrical Enclosures" or current equivalent ISO/EN/DIN/JIS/IEC standards for arc flash safety. At the time of proposal, the vendor shall indicate which document they will utilize for compliance to the requirements.

3.3.15 Controller Features. The machine shall be equipped with a Fanuc Series 0i-F Plus, or greater, CNC controller incorporating all standard features. The CNC unit shall provide programmable automatic control of machine functions, operating modes, axis movement and other part program directed functions for the machine. Movement in all axes shall be both synchronous and independent positioning. Control features shall include Manual Data Input (MDI), simultaneous control of all axes, linear interpolation, circular interpolation, programmable interface, part program edit, part program storage, background editing function, fixed cycles, and control diagnostics. The CNC shall direct machine functions from command data stored in memory. The data stored in memory shall be input by the media specified herein. The control shall initiate a halt and error signal should a fault condition occur in the control. The control shall automatically shut down when internal operating temperatures exceed safe operating limits. The controller shall comply with SAE EIA-274-D. Media Access Control (MAC) addresses for all equipment shall be provided at least 45 days prior to delivery.

3.3.15.1 **<u>Pendant Control.</u>** The machine shall be provided with a handheld pendant control to be utilized for set-ups while the enclosure doors are open. The pendant shall consist of, but not limited to, the following controls:

A. Cycle start/stop/feed hold
B. Emergency stop
C. Selector switch – each axis
D. Jog switch with manual capability

3.3.15.2 **<u>Universal Serial Bus (USB) Connection Port.</u>** A minimum of two (2) Standard USB connection ports for the transfer of process information to the system control shall be provided. All protocol documentation shall be supplied.

3.3.15.3 **<u>Ethernet Connection Port.</u>** An RJ-45 connection port for Ethernet shall be readily available on the exterior of the machine. The CNC controls shall have the capability to upload and download part program results using standard Ethernet protocol.

3.3.15.4 **<u>CNC and Control Restoration Data.</u>** The contractor shall be responsible for the generation of all program logic and documentation necessary for machine operation. The Programmable Memory Controller (PMC) program, CNC parameters and all data required for configuring a new control and any other configured or programmed electronic device shall be provided in electronic format on read-only optical disc and delivered with the equipment. For safety and security reasons, the vendor shall not provide encrypted files containing proprietary cycles PMC, PLC programs/data.

3.3.15.5 **<u>GibbsCAM Post Processor.</u>** A GibbsCAM post processor proven by third party shall be provided with the machine. The contractor must acquire a GibbsCAM post processor that is fully compatible with the computer aided drawing/computer aided machining (CAD/CAM) system, to be delivered with the machine for creation of all machine programs. The processor shall produce working programs for the machine tool required in the document from drawings created in the CAD/CAM system. The machine simulation software shall be included for safe testing of the program before execution. All necessary Virtual Machine Manager (VMM) and Machine Definition Document (MDD) files shall be included with the post processor. Testing and Debugging of the post processor is the sole responsibility of the contractor. The post processor shall utilize all functions of the equipment (i.e. simultaneously utilizing all axes for turning, chamfering, machining, boring, facing, drilling, tapping, contouring, etc.). The contractor will utilize the post processor to create the machine program for the tests. Any and all damage from the testing/prove out of the post processor shall be the sole responsibility of the contractor and all repairs shall be made by the contractor at their expense. This working post processor shall become part of the final acceptance testing and be witnessed by the Government. The final acceptance shall not be completed until a fully functioning post processor is demonstrated.

3.3.15.6 **<u>Laptop for Offline Programming.</u>** A laptop shall be provided with the following characteristics:

a. Dell Pro Max 18 Plus
b. Windows 11 Pro English 23H2
c. Processor: 32-Core 2.2 GHz 13th Gen Intel Core i9-13950HX Intel64 Family 6 Model 183 Stepping 1 Processor speed 2200 (Min)
d. Memory: 64GB, 1x32GB NECC 4800MHz DDR5 CAMM Module (min)
e. Video Graphics Card: NVIDIA RTX 5000 Ada Generation Laptop GPU (min)
f. Hard Drive: M.2 2280 1TB, Gen 4 PCIe x4 NVMe, Solid State Drive
g. Optical Readers shall be external
h. Intel Wi-Fi 6E AX210 Wireless Card with Bluetooth 5.2
i. Wireless Driver: Intel Wi-Fi 6E AX211 160MHz
j. Intel Ethernet Connection (22) I219-LM
k. TPM Module (Trusted Platform Module 2.0)
l. 110 Volt Power Cord
m. 1 Year onsite service with 1 Year NBD Limited Onsite Service After Remote Diagnosis
n. Palm Rest, with Smart card only (Microsoft Usbccid Smartcard Reader (WUDF))
o. 17.3" Anti Glare Laptop Screen with Camera and Mic
p. CAT 6 cable connection to equipment, computer, and facility local area network, all 4 connections simultaneously active.
q. Shall be TAA compliant.
r. MAC address for laptop shall be provided to customer at least 60 days prior to delivery.

3.3.15.7 **<u>Accessories for Laptop.</u>** The following laptop accessories shall be provided:

a. Two (2) Dell UltraSharp 24 InfinityEdge Monitors - U2422H
b. Dell MDS19 Dual Monitor Stand
c. 240W AC Adapter
d. Dell Smartcard Keyboard KB813/ with corded mouse.
e. Dell Performance Dock - WD19DC 240W PD

3.3.16 **<u>Motors.</u>** Motors shall be rated for continuous duty. Alternating-current (AC) motors shall be designed to operate on 60-Hz and each motor enclosure shall meet the requirements for a drip-proof enclosure. All motors shall meet the requirements of NEMA MG-1 or current equivalent ISO/EN/DIN/JIS/IEC standards. At the time of proposal, the vendor shall indicate which document they will utilize for compliance to the requirements. Each motor shall have an identification plate in accordance with paragraph 3.7.2 to identify manufacturer, model number, serial number, voltage, amperage, horsepower, phase, frequency, and frame number. Motor horsepower shall be the manufacturer’s standard or standard optional horsepower as indicated on the manufacturer's design and construction drawing(s).

3.3.17 Work light. A work light, whether built-in or added on, shall be provided. The light shall comply with NFPA 79 requirements, have a protective shield, and provide maximum light to the work area.

3.3.18 Facility Connections. Electrical connections made to the facility electrical system shall conform to NFPA 70.

3.3.19 Electrical Outlet. One (1) 120V duplex standard Type B receptacle shall be supplied at the operator station.

3.3.20 Hour Meter. The machine shall be equipped with a non-resetting type, hour meter to display accumulated operating time of the drive motor(s), mounted on the main electrical panel visible from the outside, sealed to prevent the entrance of dust and moisture, and shall be mounted to withstand shock and vibration generated by the machine. The meter shall have a range of 0 to 99,999.9 hours in increments of one (1) tenth of one (1) hour. A CNC control that tracks and displays total machine on time (a non-resetting type), controller on, program runtime, and run time of the servo motor(s) shall be considered to meet this requirement. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.4 Size and Capacity. The size and capacity of the machine shall be not less than specified in Table I.

Table I: VTL Size and Capacity

Max Allowable Footprint	21foot 6 inches x 18 foot 8 inches
Max Allowable Height	16 foot 6 inches
Bridge Crane Rail Height	14 foot 6 inches
Crane Tip Height	12 foot 1-1/2 inches
Minimum Table Diameter	62 inches
Minimum Swing Diameter	72 inches
Machining Diameter	72 inches
Machining Height	49 inches
Table RPM	250 rpm
Table Power (S1, 100%)	75 hp
Table Capacity	2,000 pounds
X & Z Feed Rates	550 inches per min
X & Z Traverse Rates	550 inches per min
X & Z Guideways	Hydrostatic Bearings
Y Axis Milling Capability	Virtual

3.5 Alignment and Accuracy Tolerances. The alignment and accuracy tolerances for the machine shall be as specified in Table II.

Table II

Position	X	± 0.00040 inches
	Z	± 0.00040 inches
	C	± 6 arc seconds
Repeatability	X	± 0.00028 inches
	Z	± 0.00028 inches
	C	± 4 arc seconds

* Machine's alignment and accuracy tolerances shall tested and measured in accordance with ISO 230-2 & ISO 230-7.

3.6 <u>Standard Tooling and Equipment.</u> All standard tooling and equipment required for proper operation and maintenance of the machine shall be provided. A list of these items shall be furnished at time of proposal.

3.6.1 <u>Tooling and/or Required Accessories.</u> <u>Tooling and/or Required Accessories.</u> Tooling in accordance with Table III shall be provided with the machine. All tooling below shall have a Capto C8 tool-holding interface from Sandvik Coromant brand, or equivalent high-performance, modular, quick-change tooling system designed for heavy-duty turning, milling, and drilling. Coolant shall be able to be used with all below tooling.

<u>Table III</u>

CNMG432 180mm GL T MAX P Cutting Unit	C8-DCLNL-55080-12	2
CNMG432 Coromant Capto extension adapter	C8-391.01-80 100A	4
DNMG432 180mm GL T MAX P Cutting Unit	C8-DDUNL-55080-15	2
Dampened 329mm GL Coromant Capto® to CoroTurn® SL damped adaptor	C8-570-3C 50 297-40L	2
Dampened 500 mm GL Coromant Capto® to CoroTurn® SL damped adaptor	C8-570-3C 60 468-40L	2
CNMG432 head T-Max® P head for turning	570-DCLNL-40-12-L	4
CNMG542 T-Max® P head for turning	570-DCLNL-40-19-L	4
DNMG432 T-Max® P head for turning	570-DDUNL-40-15	4
Optional 10" length T-Max® P cutting unit for turning	C8-DCLNL-55080-12	2
280mm GL Coromant Capto® extension adaptor	C8-391.01-80 200	2
Optional 10" length T-Max® P cutting unit for turning	C8-DDUNL-55080-15	2
DNMG432 180mm Coromant Capto® extension adaptor	C8-391.01-80 100A	2

Capco Tool Holder	C8-ASHL45-50135-20-A	4
Top Lok, shank tool for parting and grooving	TLSR-203D	4

3.6.2 **<u>Tooling Storage.</u>** Vidmar, or equivalent, tooling cabinets for storage of all required tooling shall be provided. Tooling shall be precisely shadowed within each container to support the ability to instantly see which tools are missing. Additionally, a quantity of two (2) Vidmar, or equivalent, counter-height modular drawer cabinets with five (5) drawers and overall dimension of 30-inch wide by 27.75-inch deep by 37-inch high shall be provided. All cabinets shall be fitted with four inch castor wheel, two wheels being swivel type and two wheels having a fixed position.

3.6.3 **<u>Workstation.</u>** One (1) Vidmar Type A-1, or equivalent, straight workstation with one (1) 1-SEP1023AL cabinet, one (1) BL1751 bench leg, and one (1) HT60 hardwood top shall be provided.

3.7 **<u>Marking on Instruments, Control Panels, and Plates.</u>** All words on plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. All plates shall be corrosion resistant.

3.7.1 **<u>Lubrication Plate.</u>** A lubrication plate shall be permanently and securely attached to the machine. The plate shall contain the following information:

a) Points of Lubricant Application
b) Servicing Interval
c) Type of Lubricant
d) Viscosity
e) Military or Federal Specification for Each Lubricant (If available)
f) Size and Type of Each Lubricant Filter

3.7.2 **<u>Nameplate.</u>** A nameplate shall be permanently and securely attached to the machine. If the machine is a special model, the model designation shall include the model number of the basic standard machine and a suffix identified in the manufacturer's permanent records. The nameplate shall contain the information listed below.

- Nomenclature
- CAGE Code
- Manufacturer’s Name
- Manufacturer’s Model Designation
- Manufacturer’s Serial Number
- Power Input (Volts, Total Amps, Phase, Frequency)
- Contract Number
- Date of Manufacture
- Full-load current for each supply
- Ampere rating of the largest motor or load
- Short-circuit current rating of the industrial control panel

- Max ampere rating of short-circuit and ground fault protective device, where provided
- Electrical diagram number(s) or number of the index to the electrical drawings

3.7.3 <u>Identification Marking of Military Property.</u> An Item Unique Identification Marking (IUID) shall be provided in accordance with MIL-STD-130N and all applicable documents within the standard with Machine Readable Information (MRI) for item identification marking and automatic data capture. The application of Human Readable Information (HRI) shall be used in combination with MRI and free text. At a minimum this tag shall contain the information listed in paragraph 3.7.2 Nameplate.

3.8 <u>Technical Data.</u> Three (3) hard copies and two (2) electronic copy(s), on compact disk (CD), of technical data shall be provided with the machine. Technical data shall contain specifications, instructions for operation, maintenance of the machine and recommended spare parts list. All copies of technical data shall be legible and written in the English language. Technical Data shall include the following:

a) All engineering drawings shall be in accordance with ASME Y14.100. Technical data shall contain technical information, specifications, instructions for the operation, maintenance of the machine, consumables and expendables information (such as oils, fluids, grease, filters…), and recommended spare parts list.

b) Factory service and operator manuals for the type and model of the machine being purchased. Manuals shall include electrical/electronic, pneumatic, and hydraulic schematics for the machine.

c) Diagnostic procedures for isolating and correcting faults. Procedures shall be included in the service manuals for recalibrating and performing operational checks to ensure proper operation of any replaced parts, as required.

d) Preventative and predictive maintenance tasks, including oil and vibration analysis, and show recommended intervals of tasks.

e) Software manuals and registration keys/codes shall be provided, along with original software installation disks.

f) The Contractor shall provide the Government a copy of all test results completed in section 4.7 of this specification.

3.8.1 <u>Plan of Action and Milestones (POA&M).</u> The contractor shall provide a POA&M to identify contractor performance dates of all major events from contract award to completion of all contract requirements. The initial POA&M shall be delivered thirty (30) calendar days After Contract Award (ACA). The milestones shall be prepared using a recognized project management technique (preferably Microsoft Project). The Government and the contractor shall review the report on a monthly basis. The contractor shall notify the Government prior to making any changes to the POA&M. All changes to the POA&M

shall be approved in advance by the Government. The POA&M shall detail such items as system design concept, system development, software design concept, software development, engineering, procurement of major hardware, assembly, operational debug, factory acceptance, shipment, installation, training, and final acceptance test.

3.8.2 Acceptance Test Plan (ATP). The Contractor shall prepare an Acceptance Test Plan (ATP). The ATP shall be a step-by-step written instruction on how to verify compliance during Contract Quality Assurance Inspection and the Final Inspection/Acceptance Test specified in Sections 4.4 and 4.5. The Contractor shall develop and provide dimensioned drawings of the proposed test part specified in Paragraph 4.7.3. The Manufacture's standard testing procedures shall be included in the ATP. Any procedures necessary for compliance with Section 4, which are not covered by the Manufacturer's standard testing procedures, shall be identified and included in the ATP. The initial ATP shall be submitted thirty (30) calendar days ACA for Government review and approval. The accepted test procedures shall become part of this contract.

3.8.3 Test Part Model and Drawings. The Contractor shall develop a three-dimensional (3D) test part model (igis, stp, solid, or parasolid) for the performance and post processor testing outlined in section 4.7. The test parts shall be in accordance with ISO 13041-6 (Contractor shall use category 3 size range located in ISO 13041-6, and include all tests applicable to the OEM machine working envelope stated in this SOW). The Contractor shall submit the test part model and drawing along with the POA&M and ATP. The post processor test part shall utilize all axes, functions, and features of the machine. The Government will review the test part model within fifteen (15) business days of submission. The Contractor shall re-submit any required changes to the test part model within five (5) business days for review and approval by the Government. The Contractor shall provide the Government finalized dimensioned drawings of the approved test part sixty (60) calendar days after award of contract. Test part model drawings shall be per ASME Y14.5.

3.8.4 Control of Hazardous Energy (Lockout/Tagout) Procedures. A written procedure for isolation of all energy sources on all equipment provided under this contract, in accordance with 29 CFR 1910.147, shall be provided by the contractor and included in the technical manuals. The draft procedures shall be submitted in Microsoft Word compatible format thirty (30) calendar days ACA for Government review and approval.

3.8.5 Equipment Drawings. The Contractors proposal shall be provided with dimensioned equipment drawings in both a .PDF file and CAD file compatible with AutoCAD 2015 edition. The dimensioned CAD file shall be drawn to actual size and show footprint and any manufacturer's special foundation requirements of new equipment to include ancillary pieces, utility connection points, access doors and required maintenance clearance dimensions. The CAD file shall show side and elevation views.

3.8.5.1 Installation/Foundation Drawings. The drawings shall contain at least, but shall not be limited to, the following information:

- Overall and principal dimensions in sufficient detail to establish limits of space in all directions required for installation, operation and servicing. See Enclosure 1 for maximum allowable footprint.

- Amount of clearance required to permit opening of doors and removal of plug-in units
- Clearance for travel or rotation of any moving parts
- Interface mounting and mating information, such as dimensions of location for attaching hardware, and meet the manufacturer vibration design requirement
- Interface pipe and cable attachments required for installation and co-functioning of the item to be installed with related items
- Mounting place details, hardware requirement lists, drilling plans and shock mounting buffers
- Weight of each component and total unit
- Location, type and dimensions of cable entrances, terminal boards, electrical connectors, and ducts
- Interconnecting and cable detail
- Peak electrical load (Amps) requirement to support all proposed equipment in its fully configured and operational state, including all accessories

3.8.5.2 Installation/Foundation Drawings (Preliminary Submission). Not later than sixty (60) calendar days after effective date of contract, installation drawings shall be provided with soft metric measurements using the manufacturer's standard format. An electronic copy shall also be provided in DWG or DXF format in the receiving activity's version of AutoCAD. The Government shall notify the contractor in writing within 15 calendar days of receipt of drawing if they comply with this specification. If revisions are required, the contractor has 15 calendar days to make revisions and resubmit for a new review. Rejection of drawings does not relieve the contractor from final delivery date of contract performance period.

3.8.5.3 Installation/Foundation (Final submission). Submission of final installation drawings shall be not later than sixty (60) calendar days of Government acceptance of preliminary drawing. These final drawings shall bear a complete title block with a permanent drawing number, signature, date, and an original stamp or seal of a Professional Engineer (PE). The final drawing submission shall include an engineering report with structural calculations used in the design. The engineering report shall also be stamped by a PE. The Government shall notify the contractor in writing within 15 calendar days of receipt of drawing if they comply with this specification. If revisions are required, the contractor has 15 calendar days to make revisions and resubmit for a new review. Rejection of drawings does not relieve the contractor from final delivery date of contract performance period.

3.8.6 Instrument and Calibration Procedures. Calibration procedures and/or tooling requiring operator verification or recalibration as part of a production run or on a daily or periodic basis shall be provided to the Government. The contractor shall provide a list of these items at the time of proposal. All items shall have a certificate of calibration traceable to the National Institute of Standards and Technology (NIST). The contractor shall submit draft procedures thirty (30) calendar days ACA for Government review and approval.

3.9 Installation. Contractor shall be responsible for the Installation Responsibilities sheet specified herein.

3.9.1 **Rigging.** The contractor shall be responsible for all equipment (forklifts, man lifts, etc.) and qualified personnel for unloading, rigging, installation, anchoring, connecting machine to utilities, and leveling the machines in place. The contractor shall provide the necessary anchor bolts, leveling screws, and related hardware to secure the machines' components to a foundation.

3.9.2 **Field Supervisor.** The Contractor shall provide a full-time field supervisor to direct installation and testing of the supplies from start of installation to final acceptance. The field supervisor shall be capable of reading, writing, and conversing fluently in the English language. The field supervisor shall have full authority to implement field decisions expeditiously. No work shall be accomplished when the field supervisor is not in the immediate work area. The field supervisor shall be identified to the receiving activity prior to start of installation. The field supervisor shall be responsible for coordinating with the receiving activity for any security clearances for all contractor personnel expected to perform on-site work. The receiving activity Government Point of Contact shall be notified of any change in personnel performing this duty.

3.9.3 **Installation Plan.** The Contractor shall be responsible for providing an Installation Plan. The Installation Plan shall contain detailed information regarding all installation work. The initial Installation Plan shall be delivered to the Government within thirty (30) calendar days After Contract Award (ACA) for review and approval.

3.9.4 **Compressed Air Connection.** The Contractor shall be responsible for design and installation of new compressed air piping and associated components. The existing compressed air pressure available at the Government facility fluctuates between 83 to 93 psi. The contractor shall assume that all compressed air qualities to be of the lowest quality that is useable as compressed air. The Contractor shall make provisions to filter, dry, humidify, lubricate, amplify, and regulate the Government supplied compressed air to meet the Original Equipment Manufacturer (OEM) compressed air requirements and recommendations. A fully functioning chip blow gun shall be provided near the operator's work's station with reach to the extents of the work zone

3.9.5 **Existing Equipment and Utilities Removal.** The contractor shall be responsible for removing replaced equipment and utilities. The existing piece of equipment to be removed is the Bullard Dynatrol 66" Vertical Turret Lathe (asset: 65887603571) located in Building 133. The removal of utilities shall include, but not be limited to, associated components, piping, conduit, and wiring. The Contractor shall lock out the machine one (1) week prior to its removal to allow Government personnel to perform initial decontamination of the machine. The Contractor shall then transport the equipment to Building 4466 for final decontamination ~~and disposal~~ by the Government. Once decontamination is complete, the Contractor shall transport the equipment to DLA Disposition Services. Estimated travel is two (2) miles within the confines of the facility.

3.9.6 **Geotechnical Survey.** The contractor shall perform a geotechnical survey to determine if and where there are any hidden utilities in the location of the current foundation. The onsite survey shall be completed within thirty (30) calendar days ACA, coordinate with the Contracting Officer.

3.9.7 Machine Foundation. The Contractor shall be responsible for providing a machine foundation conforming to Original Equipment Manufacturer (OEM) requirements. For unforeseen issues that are found during installation of the foundations, the contractor shall notify the facility point of contact (POC) and the Contracting Officer immediately to determine further course of action.

3.9.8 Vibration Isolation System. The machine shall be installed with a vibration isolation system.

3.9.9 Telecommunication Connection. The Contractor shall be responsible for coordinating with the FRCE Project Manager and IT group via DLA contracting officer regarding the final connection and completion of active communication signals of equipment along with any required stand-alone computer to the Government facility Distributed Numerical Control (DNC) system.

3.9.10 Lead and Asbestos Testing. The Contractor shall be responsible for conducting an onsite survey for Lead Based Paint (LBP) and Asbestos Containing Material (ACM). The onsite survey shall be completed within thirty (30) calendar days ACA, coordinate with the Contracting Officer. The facility has a known presence of LBP and ACM. A North Carolina accredited Asbestos Inspector shall perform the onsite survey for Lead and Asbestos to identify any suspected material that appears to be involved in the project. Assume the presence of asbestos and lead paint is a possibility and be prepared to do some sampling and analysis. Provide a signed report certifying the inspection. State the findings of the inspection and state if the area appears to be asbestos free or present the results of the sampling and analysis. The report shall be certified by the Asbestos Inspector and include the Inspector's certification number. Show locations of any samples taken in the report and on the Installation Plan. If either of these two substances are found in areas that will be disturbed by the Installation, and will pose a health and safety threat, the contractor must notify the contracting officer. The contractor is not required to remove or abate the lead and asbestos under this contract.

3.9.11 Work Stoppage. For unforeseen issues that are found during installation, the contractor shall notify the facility point of contact (POC) and DLA immediately to determine further course of action.

3.10 Training. After satisfactory completion of acceptance testing of the system, the services of a qualified representative(s), who is proficient in the English language, shall be provided for specialized training to familiarize Government personnel with the equipment and to help ensure reliable performance and maximum service life, during normal usage. The course shall utilize as-built drawings and manuals provided under this contract and shall include both classroom and hands-on portions. All printed training aids ~~shall be~~ supplied for the course shall be in the English language and become the property of the Government.

3.10.1 Operator Training. Operator training shall be provided at FRC East. Training shall be provided for a minimum of five (5) operators. Training shall not be less than two (2) weeks of four (4) consecutive eight (8) hour workdays, between 6:30 am and 3:00 pm excluding weekends and federal holidays. Training shall include preparation of equipment for safe operation and proper isolation of equipment procedures. The operator shall be provided instruction for the care, use and operation of all attachments and accessories. Training manuals shall be provided for each student.

3.10.2 GibbsCAM Post Processor Programmer Training. GibbsCAM programmer training shall be provided at FRC East facility after satisfactory completion of Operator Training. The training shall be for a minimum of three (3) personnel. Training shall not be less than three (3) consecutive eight (8) hour workdays, between 6:30 am and 3:00 pm excluding weekends and federal holidays. The contractor shall provide all training supplies for the training. The Contractor shall provide laptop training computers and training manuals to each student. Programmer training shall consist of training utilizing all features and modules of the software package specified in paragraph 3.3.15.5. The training shall utilize the laptop specified in 3.3.15.6.

3.10.3 OEM Facility Maintenance Training. Mechanical and Electronic/Electrical maintenance training shall be provided for six (6) total maintenance personnel. The training location shall be performed at a certified training facility of the machines manufacturer and shall not be less than five (5) consecutive eight (8) hour workdays. Training manuals shall be provided for each student. The maintenance course shall cover at a minimum:

a) Fanuc Controller and Machine Overview (Basic overview of anything new with FANUC)
b) Fanuc Diagnostics
c) Backing up SRAM, parameters, and event logfile
d) Review of systems drawings and schematics
e) Electrical Troubleshooting
f) Mechanical Troubleshooting
g) Instruction manual and how to navigate
h) Location and replacement of all belts
i) Explanation of coolant, lubrication, and hydraulic systems
j) Recommended fluids and common spare parts
k) Overview of wheel head & work head spindles and assemblies
l) Preventative maintenance overview weekly, monthly, annual PM
m) Checking and correcting alignment accuracy– review of required tools
n) Review questions to finalize session.

3.10.4 Maintenance Training. Mechanical and Electronic/Electrical maintenance training shall be provided for six (6) total maintenance personnel. Training shall be performed at the Government facility and shall not be less than four (4) consecutive eight (8) hour workdays each, between 6:30 am and 3:00 pm excluding weekends and federal holidays. Training manuals shall be provided for each student. The maintenance course shall cover at a minimum:

a. review of systems drawings and schematics
b. component location and function
c. troubleshooting procedures and techniques
d. repair procedures including disassembly and assembly

e. system maintenance and servicing
f. setup, adjustments, and calibration (how, when, and where)
g. preventative maintenance procedures
h. Procedures for proper isolation.
i. Basic start up, manual, auto, and MDI mode
j. SRAM, Ladder, PMC Memory, Parameters, Backup and Restoring

4.0 QUALITY ASSURANCE.

4.1 Responsibility for Inspection. The contractor shall be responsible for the performance of all inspection requirements specified herein and in accordance with the attached Quality Assurance Provision (QAP). Except as otherwise specified in the contract or purchase order, the contractor may use his own facility for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to witness any inspections set forth in the purchase description where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for Compliance. The Contractor shall be responsible for ensuring all items meet the requirements of sections 3 and 4.

4.2 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Contract quality assurance inspection (Origin) (see 4.4).

b. Final Inspection/Acceptance test (Destination) (see 4.5).

4.3 Inspection Conditions. All inspections, tests, and examinations shall be performed in an indoor facility with ambient temperature conditions in the range of 55° F to 100° F and 10% to 95% relative humidity.

4.4 Contract Quality Assurance Inspection (Origin). Contract quality assurance inspection shall be applied at the contractor's facility prior to being offered for acceptance under the contract. The machine and all peripheral equipment added to the machine shall be fully assembled and perform all functions in accordance with this specification. Quality conformance inspection shall consist of the examination in 4.6 and the tests in 4.7. The machine shall pass the examination, all tests, and contract quality assurance inspection to be acceptable for shipment.

4.4.1 Exception to Contract Quality Assurance Inspection (Origin). The application test in paragraph 4.7.6 shall be performed at destination only.

4.5 Final Inspection/Acceptance Test (Destination). Final Inspection and acceptance test shall be performed on the machine to ensure conformance with this purchase description. The acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination in 4.6 and all tests in 4.7. The machine shall pass the examination and all tests to be accepted.

4.5.1 Exception to Final Inspection/Acceptance Test (Destination). The horsepower test in paragraph 4.7.5 shall be performed at origin inspection only.

4.6 Examination. The equipment shall be examined for design, dimensions, construction, materials, components, electrical equipment and workmanship to determine compliance with the requirements of this specification.

4.7 Tests. Unless otherwise specified within this document, all programs, instruments, materials, material handling equipment, qualified personnel, and tools required to perform and evaluate the tests shall be furnished by the contractor. All measurement tools and indicating devices shall be calibrated in accordance with the attached QAP. If a malfunction occurs during the test, the test shall be restarted each time until all tests in 4.7 are completed without a malfunction. Three (3) consecutive failures without completion of the test may be considered cause for rejection of the system. For the purpose of this test, a "failure" is defined as any equipment malfunction which requires remedial action to restore the system to full operation in accordance with this specification.

4.7.1 Alignment and Accuracy Tests. Laser alignment tests shall be conducted to ensure the machine meets the alignment and accuracy tolerances specified in paragraph 3.5, Table II. The Contractor shall provide a report showing the origin and destination alignment and accuracy results. The machine's repeatability and accuracy for all slide motions, spindle speed changes, feed rate changes and auxiliary functions shall be checked.

4.7.2 Operational Test. The machine shall be operated under CNC at no-load for not less than four (4) hours. During operational testing proper operation of all control functions, machine response to commands, proper operation of motors, adjusting mechanisms, and auxiliary functions shall be checked. The machine shall be run in each mode of operation (Manual and Automatic) to test the machine's response to command data. Three (3) failures in a row to complete the run test may be cause for rejection of the entire system. After successful completion of the operational test, proper operation of all controls, motors, adjusting mechanisms, over-travel limits (programmed stops and hard stops), emergency stop devices, and accessories shall be verified. The safety limit stops shall be checked by attempting to exceed the limits of travel of each moving axis.

4.7.3 Performance Test. The machine shall be tested in accordance with ISO 13041-6 for the performance test. Two identical parts shall be machined for each cutting test to check dimensional accuracies, repeatability, and surface finish capabilities. The Contractor shall use the Government approved 3D test part model specified in paragraph 3.8.3. The material shall be 6061 T6 Aluminum or equivalent. Surface finish on all cuts shall be 125 to 63 µin Ra. At Origin Testing, the Contactor shall measure all part dimensions using a certified Coordinate Measuring Machine (CMM). At Final Inspection, the FRC East shall measure all part dimensions using a certified FRC East owned CMM. The finished test parts shall meet the Geometric Dimensioning & Tolerancing (GD&T) requirements in Table IV. Dimensioning and Tolerancing shall be per ASME Y14.5. At the time of proposal, the vendor shall indicate which document they will utilize for compliance to the requirements.

4.7.4 <u>Post Processor Test.</u> The machine shall be tested to ensure the post processor complies with paragraph 3.3.15.5. The Contractor shall use the Government approved 3D test part model specified in paragraph 3.8.5. Post processor testing shall begin at Origin Testing and conclude at Destination Testing. The Contractor shall allow a minimum of forty (40) hours for testing of post processor at destination. The Government shall not complete final acceptance until a fully functioning post processor has been proven. To prove out post processor, the Contractor shall:

a. Import Government approved 3D part model into Gibbs CAD/CAM software installed on Laptop.
b. Create tool paths of part model using machine simulation and material removal modules.
c. Convert cutter location code, using CAD/CAM post processor on Laptop, to usable G-code file.
d. Transfer G-code file from Laptop to machine controller. At Final Inspection/Acceptance Test the G-code file shall be transferred from supplied Laptop to RDT&E manufacturing network, then downloaded to machine controller.
e. Run G-code file on machine without errors nor manual manipulation of posted code.

4.7.5 <u>Horsepower Test. (Origin Only).</u> A horsepower test shall be accomplished and shall meet the requirements in paragraph 3.4, Table I.

4.7.6 <u>Application Test (Destination Only).</u> Turning and chamfering operations shall be performed on a scrapped (insert part here). The test shall utilize the supplied Reinshaw part and tool probes. The (insert part here) material is (insert material here). The features will be chosen by the Government and will range from (insert size, finish, etc. of feature).

4.7.7 <u>Audible Noise Level Test.</u> The equipment shall be tested for compliance with paragraph 3.1.4. “Audible Noise Level”.

INSTALLATION RESPONSIBILITIES

Government	Contractor	
☐	☒	a. Provide machine foundation in accordance with the Manufacturer’s design requirements
☐	☒	b. Furnish labor and material handling equipment for off-loading and placing item on foundation
☐	☒	c. Provide and install anchor bolts and nuts
☐	☒	d. Set and rough level the machine on its foundation
☐	☒	e. Level and align machine
☐	☒	f. Connect machine to the provided utilities hook-up
☐	☒	g. Provide all necessary tools, gages, and instrumentation necessary to perform the required tests
☐	☒	h. Provide and charge all systems with fluids in accordance with manufacturer's instructions
☐	☒	i. Provide and charge coolant system with HOCUT 795B
☐	☒	j. Material for performance testing at Government facility
☐	☒	k. Material for performance testing at Contractor's facility
☐	☒	l. Material handling, Rigging equipment and Qualified Personnel for performance testing

GOVERNMENT RESPONSIBILITIES

☒ Provide utilities hook-up within 50 feet of the machine

☐ Government will connect utilities to the machine based on the contractor's accepted site drawings.

☐ Assign one mechanic (mechanical/hydraulic) and one maintenance mechanic (electrical/electronic) to provide information and any additional materials for the contractor's representative during installation, verification, and initial start-up of the machine.

☐ No work shall be performed by customer personnel when the contractor's representative is not in the immediate work area during installation, verification, and initial start-up of the machine.